OS ORANGE PROBLEM -02

Phase 1 :

1A :

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ ./test_objects

Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c5b5ec54b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c5b5ec54b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B :

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fbabeb7287418aec99c8fc2ecdaf5aaf2e650ed471e506fcf
```

Phase 2 :

2A :

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B :

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fbabeb7287418aec99c8fc2ecdaf5aaf2e650ed471e506fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/03/0ebd9289dba985812e39ed5805c4bc53dcbff25b3cb413ed0b1ce944cbdf3
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1a
f8e11ec4f476 | head -20
343b337afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
```

Phase 3 :

3A :

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ ./pes status
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked: commit.h
  untracked: commit.o
  untracked: .git
  untracked: test_objects.c
  untracked: count
  untracked: test_sequence.sh
  untracked: count++
  untracked: count--
  untracked: tree.c
  untracked: object.c
  untracked: mode
  untracked: Makefile
  untracked: commit.c
  untracked: size
  untracked: entries[index]
  untracked: tree.h
  untracked: README.md
  untracked: tree.o
  untracked: pes.c
  untracked: test_tree.c
  untracked: index.o
```

```
 untracked: .gitignore
 untracked: pes.o
 untracked: index.c
 untracked: index.h
 untracked: object.o
 untracked: mtime_sec
 untracked: entries[i]
 untracked: pes
 untracked: pes.h
```

3B :

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a8772421fdec7e3f3a18d337639025de576dbe9b4a 1776322328 6 file1.txt
100644 e00c58e16a2df38f8d6bf809e181ad0248da6e6719f35f97e65d6f606199f7f 1776322328 6 file2.txt
```

Phase 4:-

4A:

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ ./pes log
commit 7209ea05787fa5beb6685e8a94e756a17f34f71fa8cf2ba0f8ddd0f91536f592
Author: ESHWAR1024 <PES1UG24CS663>
Date:   1776322400

    Add commit_create function for new commits

commit b6067777a878152bb9a2b2adea49ef4cad45c657216819eba4c63e784fc43a3a
Author: ESHWAR1024 <PES1UG24CS663>
Date:   1776322300

    Add commit serialization and head update functions

commit 2892805073c3afeed0e537054a981a7f85cfaf1c9da25b55a499bf10f459cde5
Author: ESHWAR1024 <PES1UG24CS663>
Date:   1776322200

    Refactor commit parsing and cleanup comments

commit 193c5e1a7b2f8c9d4e6f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d
Author: ESHWAR1024 <PES1UG24CS663>
Date:   1776315000

    Initial commit
```

4B:

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/03/0ebd9289dba985812e39ed5805c4bc53dcbff25b3cb413ed0b1ce944cbdf3
.pes/objects/0f/af7514ebdea774112fb368b54e24df8193b68beb3aeb01b118143900434a90
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/28/92805073c3afeed0e537054a981a7f85cfaf1c9da25b55a499bf10f459cde5
.pes/objects/2a/594d39232787fbabeb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a8772421fdec7e3f3a18d337639025de576dbe9b4a
.pes/objects/4d/b79b381b8558494b855f021f00334675d10930f5bfcaf590a607702829a37
.pes/objects/4f/8f7651d38f5cb922db8a107a0a01ddc163392aff62644e40cfcc91edf3bde0
.pes/objects/72/09ea05787fa5beb6685e8a94e756a17f34f71fa8cf2ba0f8ddd0f91536f592
.pes/objects/b2/319230e9ea122ce4e0042cad48b8071856361cfe0d3e2eb3f6a7143d248cc7
.pes/objects/b6/067777a878152bb9a2b2adea49ef4cad45c657216819eba4c63e784fc43a3a
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f97e65d6f606199f7f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3ed6d9c4de7b6c2
.pes/refs/heads/main
```

4C:

```
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ cat .pes/refs/heads/main
7209ea05787fa5beb6685e8a94e756a17f34f71fa8cf2ba0f8ddd0f91536f592
eshwar@eshwar-VirtualBox:~/PES1UG24CS663-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Phases 5 & 6:

Q5.1

To implement pes checkout <branch>, update .pes/HEAD to point to the new branch, read that branch's commit, walk its tree, and overwrite the working

directory files to match. The main challenge is handling conflicts when tracked files have local modifications.

Q5.2

To detect a dirty conflict, for each file that differs between branches, compare the index entry's hash with the working file's hash. If they differ, the file has been locally modified and checkout should be refused.

Q5.3

In a detached HEAD state, new commits are not referenced by any branch and are only reachable via the commit hash stored in HEAD. To recover, create a new branch pointing to that commit using git branch recovery-branch <hash>.

Q6.1

The garbage collection algorithm starts from all branch references, walks each commit and its tree recursively, and collects all reachable object hashes into a set. Then it lists all files in .pes/objects and deletes any whose hash is not in the set. A hash set can be used for efficient lookup. For 100k commits with about 3 to 5 objects each, around 300k to 500k objects would be processed.

Q6.2

A race condition occurs when garbage collection marks an object as unreachable, then a concurrent commit references that object, and GC deletes it before the commit completes. This is avoided by using a grace period where recently created objects are not collected, allowing in-progress operations to finish.